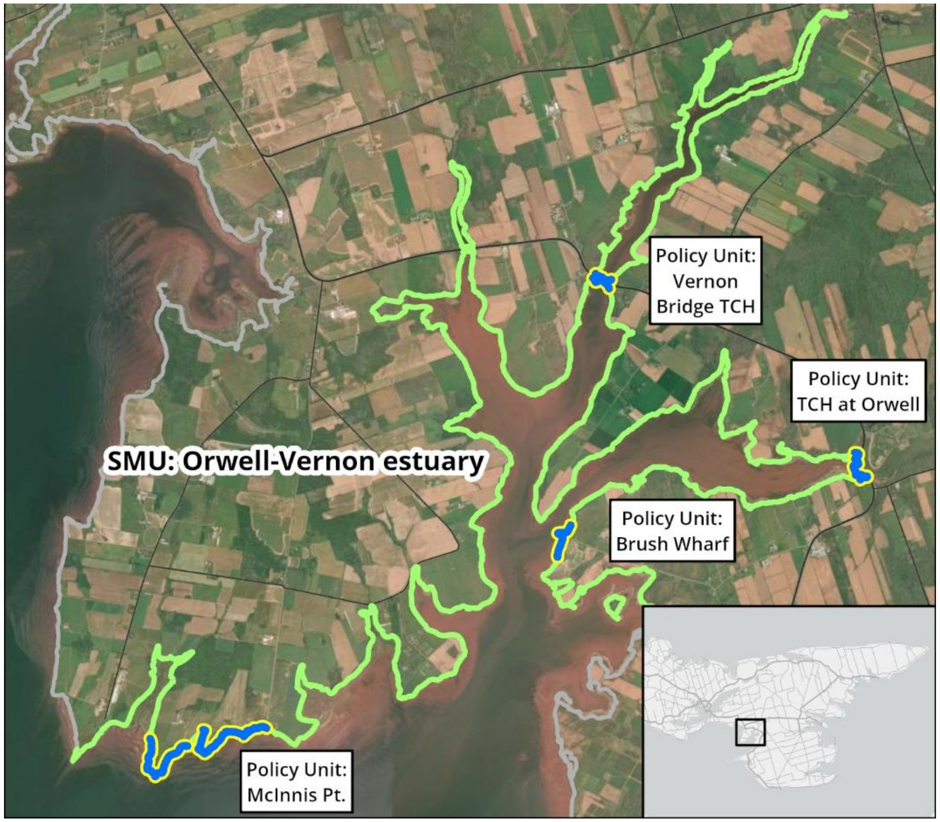
Orwell-Vernon Estuary

Shoreline length: 71 km



Shore Types

Orwell-Vernon Estuary is an estuary located on the southeast end of PEI. Shore type characterized by a majority of wetlands and low plains.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
21	4	2	3	70	0	0	0

Cliffs at McInnis Point

TCH at Vernon Bridge

Brush wharf

Low plain at the mouth of Vernon Rover estuary

Governance, Infrastructure, and Economy

This area is unincorporated.

Key infrastructure includes Orwell Corner Historic Village, Earnscliffe Road (Route 267, at the beach); Route 270 (McInnis Point Road, at a sharp turn, with potential to isolate portions of McInnis Point); Route 268 (China Point Road, at Watts and China Creeks); the TCH at Vernon Bridge; Murray Harbour Road at Vernon River; the TCH at Orwell (Uigg Road); Brush Wharf Road (Orwell Cove); and Frand Road (Orwell). Coastal economic drivers include Bill & Stanley's Oyster Company at Brush Wharf Road. Local and legacy land use constraints include several small lots, accessed by unpaved coastal roads.

Natural Assets and Heritage

The Orwell Corner Historic Village is found at the northernmost point of this SMU. Additionally, the Sir Andrew Macphail Homestead is located in this SMU.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

Infrastructure along low-lying shores are vulnerable to increasing flood hazards. Erosion hazard is limited due to the sheltered orientation of the area.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Orwell-Vernon Estuary				
Flood risk	2 [<1/km]	2 [<1/km]	6 [<1/km]	7 [<1/km]
Erosion risk	NA	17 [<1/km]	19 [<1/km]	30 [<1/km]
PU McInnis Point				
Flood risk	0 [<1/km]	0 [<1/km]	1 [<1/km]	2 [1/km]
Erosion risk	NA	9 [3/km]	11 [4/km]	17 [6/km]
PU TCH at Orwell				
Flood risk	0	0	0	0
Erosion risk	NA	1 [1/km]	1 [1/km]	1 [1/km]
PU Vernon Bridge TCH				
Flood risk	0	0	1 [2/km]	1 [2/km]
Erosion risk	NA	4 [7/km]	4 [7/km]	5 [8/km]
PU Brush Wharf				
Flood risk	1 [2/km]	1 [2/km]	1 [2/km]	1 [2/km]
Erosion risk	NA	1 [2/km]	1 [2/km]	2 [3/km]

Sediment transport - The estimated net sediment transport rate averaged across the SMU is low, except for the exposed outer cliffs near McInnis Point.

Proposed SMP Strategies

The primary strategy for Orwell-Vernon Estuary is No Active Intervention. In areas where infrastructure is at risk, managed realignment and short-term resist are options.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	**NAI**	**NAI**	**NAI**
Rationale	The local shoreline and hazard zone should remain natural, undeveloped.		
Proposed PU strategy **Mclinnis Point**	**R**	**MR**	**MR**
Rationale	A short- to medium-term Resist strategy may be implemented where necessary to protect existing buildings over the remaining lifetime.		
Proposed PU strategy **TCH at Orwell**	**MR**	**MR**	**MR**
Rationale	MR strategy should include raising critical highway infrastructure with future bridges and culverts to allow full tidal passage.		
Proposed PU strategy **Vernon Bridge TCH**	**MR**	**MR**	**MR**
Rationale	MR strategy should include raising critical highway infrastructure with future bridges and culverts to allow full tidal passage.		
Proposed PU strategy **Brush Wharf**	**R**	**MR**	**MR**
Rational	A Resist strategy is warranted for this infrastructure until the flood risk justifies MR.		

LEGEND

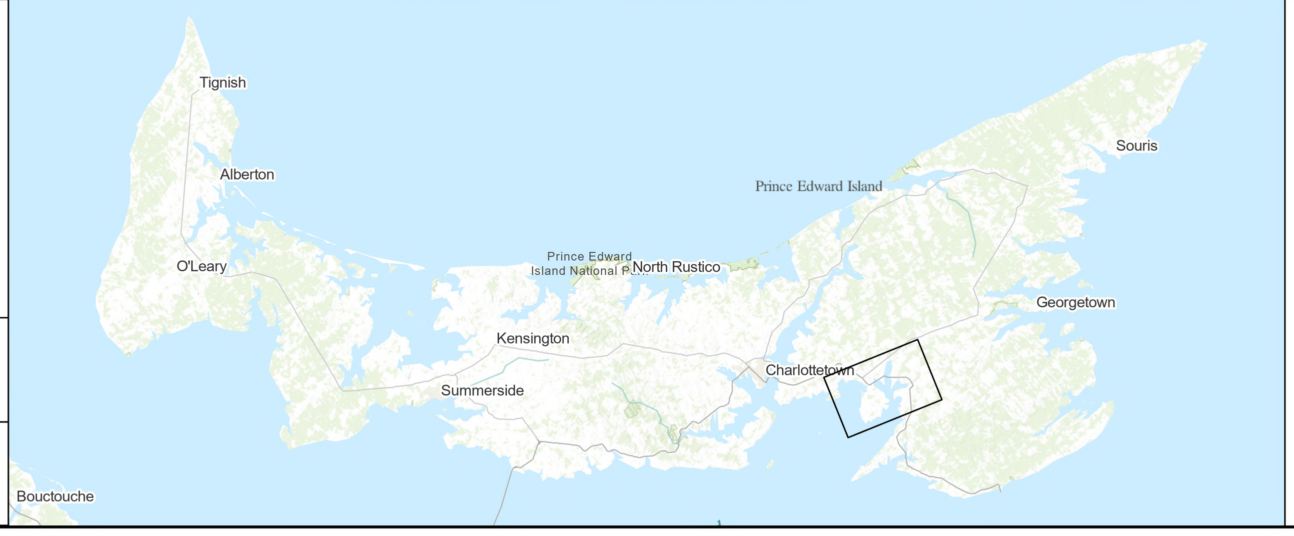
- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

N
W E
S

0 260 520 1,040
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:



COASTAL EROSION AND FLOOD HAZARD

Shoreline Management Unit: Orwell-Vernon estuary

Prince Edward Island
Île-du-Prince-Édouard

CBCL